UNIT 2 · TOPIC 6(B)

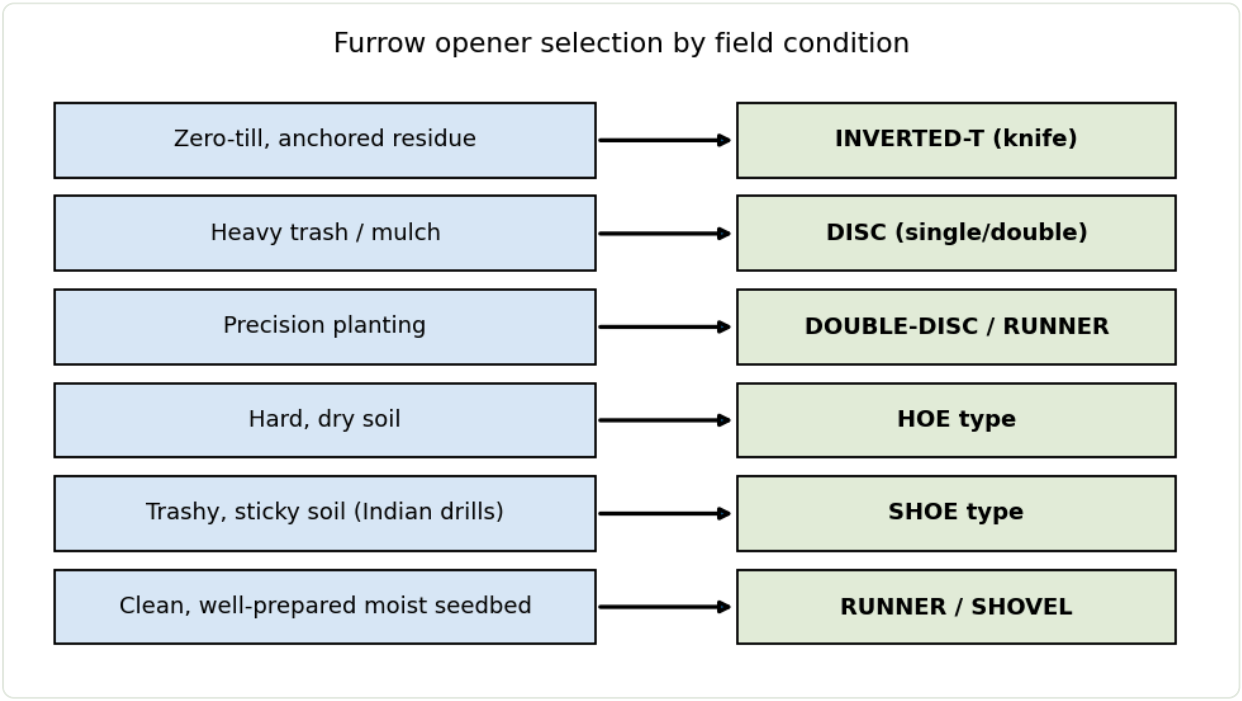
Furrow Openers, Hoppers and Seed Tubes

Syllabus topic: Design of soil working tools for sowing and planting

Furrow openers — function and types

A furrow opener opens a uniform furrow at the set depth, places seed (and fertiliser) with minimum soil disturbance, and allows proper covering. Choose it for the soil condition, moisture, residue/trash and seedbed preparation.

Opener	Use
Shovel / shoe	Most common on Indian bullock and tractor drills; the shoe works well in trashy fields and sticky soils; robust, simple.
Hoe	Hard, dry soils and rootzone-hard seedbeds; heavier draft, more disturbance.
Runner (sword/stub-runner)	Clean, well-prepared, moist seedbeds; narrow, well-formed furrow — common on planters.
Single disc / double disc	Cut through trash and residue; double-disc gives precise, uniform seed placement — standard on planters and no-till drills; needs down-pressure (springs/weight) to penetrate.
Inverted-T (knife)	Minimal-disturbance slit with a small horizontal shelf — the classic zero-till opener for sowing wheat into rice residue with anchored stubble.
Chisel/knife; winged	Deep fertiliser banding; wide-band placement.



Furrow opener selection by field condition

Opener design considerations

- **Geometry:** a small approach (rake) angle and narrow wedge angle reduce draft and disturbance; beyond an optimum rake angle, draft rises sharply. A point ahead of the seed tube protects it and firms the furrow bottom.
- **Disc openers:** design the disc diameter, disc angle (**7–10°** between the discs of a double-disc opener) and offset point.
- **Depth control:** gauge wheels/skids on each opener or the frame. Spring loading gives constant down-force on uneven ground and breakaway protection over stones (spring-loaded tines on cultivators serve the same role).
- **Down-force:** no-till openers need heavy down-force springs — **50–150 kgf per opener** for disc openers in residue; the frame and linkage must carry it.
- **Materials & wear:** points of medium/high-carbon steel, hardened or hard-faced (e.g., welded hard-facing alloy) at wear edges; replaceable/reversible points cut cost (materials in Unit 8).
- **Drill draft:** total drill draft ≈ Σ opener draft + rolling resistance; the per-metre draft of a seed drill is far lower than that of tillage implements (order **0.4–1 kN/m**).

Covering and firming

- Press wheels firm the soil around/over the seed for good seed–soil contact and moisture supply → better, faster germination. Design press-wheel load: **2–10 kgf** typical.
- Covering devices (chains, scrapers, paired press wheels) close the furrow.

Hopper and seed tube

- **Hopper:** make the sides steeper than the seed's angle of repose (wheat ~**25–28°**; design **≥ 35–40°**). Add anti-bridging agitators for fuzzy/chaffy seed. Size it for **≥ 1–2 ha** of sowing.
- **Seed tube:** short, smooth, near-vertical; spiral/telescopic tubes reduce bounce. Keep the delivery velocity low and rearward-directed to cancel the forward machine speed (reduces seed roll in the furrow).

Practice MCQs

Q103. The furrow opener best suited for zero-till sowing into anchored rice residue is:

(a) Runner
(b) Inverted-T
(c) Shovel
(d) Full-width sweep

Q104. Double-disc furrow openers are preferred on planters because they give:

(a) Maximum soil inversion
(b) Precise, uniform seed placement and cut trash
(c) Deepest furrows
(d) Lowest cost

Q105. Hoe-type furrow openers are best suited to:

(a) Wet puddled soil
(b) Hard and dry soils
(c) Loose sand only
(d) Standing water

Q106. Runner-type openers require:

(a) Trashy uneven land
(b) Clean, well-prepared moist seedbed
(c) Stony soil
(d) Deep mud

Q107. The press wheel behind a furrow opener serves to:

(a) Increase draft
(b) Firm soil around seed for better seed–soil contact and germination
(c) Meter the seed
(d) Mark the row

Q108. Disc-type openers are preferred over shoe type when:

(a) Field has heavy trash/residue
(b) Soil is clean and loose
(c) Speed is very low
(d) Seeds are very small

Q109. Increasing the rake (approach) angle of a tine well beyond its optimum causes draft to:

(a) Decrease
(b) Increase
(c) Remain constant
(d) Become zero

Q110. Spring-loaded tines on cultivators are provided mainly to:

(a) Increase working depth
(b) Protect against stones/obstructions by breakaway action
(c) Reduce cost
(d) Improve seed metering

Q122. Hopper walls must slope:

(a) Less than angle of repose
(b) Greater than the angle of repose of the material
(c) Exactly 30°
(d) Any angle

Q123. The angle between the two discs of a double-disc furrow opener is about:

(a) 2–3°
(b) 7–10°
(c) 20–25°
(d) 40°

Q124. Heavy down-force springs (50–150 kgf/opener) are characteristic of:

(a) Broadcasting
(b) No-till/zero-till openers
(c) Cup feed drills
(d) Check-row planters

Q125. A press wheel behind the opener serves to:

(a) Mark rows
(b) Firm soil around seed for better seed–soil contact
(c) Drive the meter
(d) Cut trash

Answers

Q103 — (b) Inverted-T
Q104 — (b) Precise, uniform seed placement and cut trash
Q105 — (b) Hard and dry soils
Q106 — (b) Clean, well-prepared moist seedbed
Q107 — (b) Firm soil around seed for better seed–soil contact and germination
Q108 — (a) Field has heavy trash/residue
Q109 — (b) Increase
Q110 — (b) Protect against stones/obstructions by breakaway action
Q122 — (b) Greater than the angle of repose of the material
Q123 — (b) 7–10°
Q124 — (b) No-till/zero-till openers
Q125 — (b) Firm soil around seed for better seed–soil contact

Sources: Kepner, Bainer & Barger — Principles of Farm Machinery (implement force analysis & hitching); ASAE Standard D497 (Agricultural Machinery Management Data — draft equation); Goryachkin's rational draft theory; Jagdishwar Sahay — Elements of Agricultural Engineering; NPTEL Farm Machinery & Equipment lectures. Kepner, Bainer & Barger — Principles of Farm Machinery, Ch. 11 (Crop Planting), Ch. 12 (Row-Crop Cultivation), Ch. 13 (Application of Fertilizers); standard FMP design references.